Numerical Simulation of Atmospheric Pollutant Dispersion in an Urban Street Canyon Using CFD Methods

March 26, 2026

Abstract

This report presents a numerical investigation of atmospheric pollutant dispersion in an idealized urban street canyon using Computational Fluid Dynamics (CFD). The study is based on a reference street-canyon benchmark and focuses on the transport of sulfur hexafluoride (SF_6) under steady approaching wind conditions. A steady-state Reynolds-Averaged Navier-Stokes (RANS) framework with species transport is used to simulate the airflow and pollutant concentration field. The geometry, mesh strategy, governing equations, boundary conditions, and User-Defined Functions (UDFs) are described in detail. The main objective is to identify the concentration distribution and possible pollutant accumulation zones within the canyon.

Contents

Chapter 1

Introduction

1.1 Background and Motivation

Air pollution in urban areas is an important problem for public safety and urban planning. In street canyons, the interaction between wind and buildings can reduce natural ventilation and create recirculation zones. As a result, pollutants may accumulate near pedestrian level and remain trapped for a long time in weakly ventilated regions. This makes street canyons one of the most important urban configurations for dispersion studies. Therefore, numerical simulation is useful for understanding pollutant transport and identifying poorly ventilated regions.

1.2 Problem Statement

This study considers an idealized urban street canyon with aspect ratio $W/H = 1$, where both the street width and building height are equal to 18 m. The configuration is based on a benchmark previously studied in wind-tunnel and CFD investigations. Sulfur hexafluoride (SF_6) is released at street level and transported by an approaching atmospheric boundary layer flow. The main goal is to analyze the airflow pattern and pollutant concentration distribution inside the canyon under steady inflow conditions. Special attention is given to the recirculation structure inside the canyon, since it strongly affects pollutant removal and accumulation.

1.3 Sulfur Hexafluoride (SF_6) as a Tracer Gas

In this work, sulfur hexafluoride (SF_6) is used as a tracer gas for pollutant dispersion modeling. It is released from street-level line sources that represent traffic-related emissions. This gas is widely used in dispersion studies because it allows researchers to track the transport of pollutants under controlled conditions. By analyzing the concentration field of SF_6, it is possible to evaluate ventilation performance and identify regions where pollutants tend to accumulate.

1.4 Objectives

The main objectives of this study are:

- to construct the 3D geometry of the urban street canyon;
- to implement a CFD setup in ANSYS Fluent for steady-state airflow and species transport;
- to define suitable boundary conditions for the approaching flow and pollutant release;
- to analyze the pollutant concentration distribution and identify accumulation zones.

Chapter 2

Geometry and Mesh

2.1 Geometry and Computational Domain

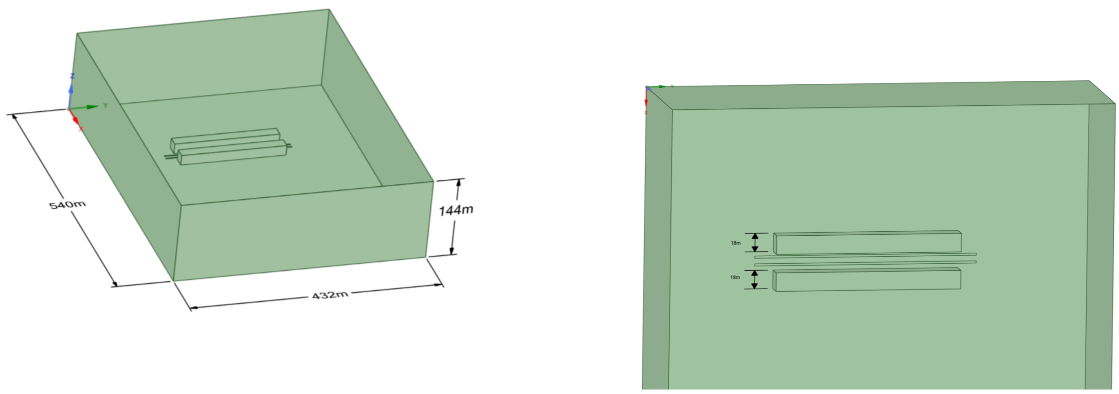
The numerical investigation is based on an idealized Urban Street Canyon (USC) configuration. The geometry is defined by a unity aspect ratio, where the street width (W) and building height (H) are both set to 18 m ($W/H = 1$). This specific configuration was chosen to maintain consistency with the experimental wind-tunnel (WT) data provided by the University of Karlsruhe, Germany.

The validation of the current numerical model relies on the comparison between computational results and concentration measurements available in the public scientific database, as discussed in previous studies comparing *Reynolds-Averaged Navier-Stokes* (RANS) and *Large Eddy Simulation* (LES) approaches for atmospheric pollutant dispersion.

The first phase of the project involved the construction of a 3D model representing the street canyon area. The flow domain and its respective boundary conditions are defined as follows:

- **Main Air Inlet:** A velocity-inlet profile representing the atmospheric boundary layer.
- **Outlet:** A pressure-outlet boundary condition with radial equilibrium pressure distribution was applied to allow the flow to leave the computational domain.
- **Pollutant Sources:** In this study, two line sources were used as a simplified representation of the benchmark traffic-emission configuration.

The detailed dimensions of the computational domain and the nested subdomains are illustrated in Figure 2.1.



(a) Top view of the domain

(b) subdomain

Figure 2.1: Computational domain setup: (a) layout and subdomains; (b) applied boundary conditions and line source locations.

2.2 Computational Mesh

To achieve a balance between computational cost and spatial resolution, a hybrid hexa-dominant mesh was generated. The domain was discretized using a multi-zone approach, in which the region surrounding the street canyon and the pollutant sources was heavily refined. The mesh parameters are summarized as follows:

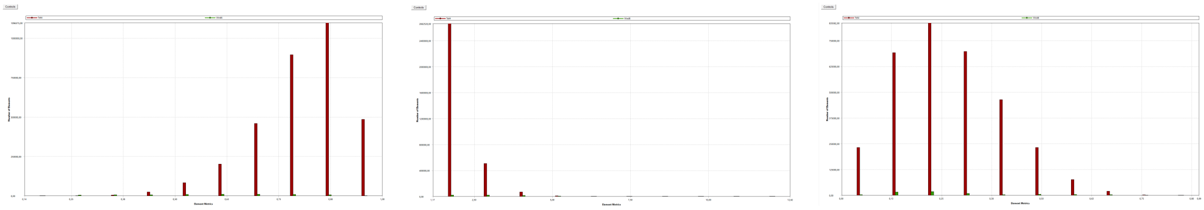
- **Global Mesh Size:** $H/4.5 \approx 4$ m, ensuring a baseline resolution for the far-field flow.
- **Refinement Zone (Sub-mesh):** Within the street canyon and near building surfaces, the element size was reduced to $H/9 \approx 2$ m.
- **Boundary Layer Treatment:** To accurately resolve the viscous sublayer and pressure-driven separation at the building corners, 5 inflation layers were applied to all solid wall boundaries.
- **Mesh Statistics:** The final discretized domain consists of approximately 591,495 nodes and 3,388,605 elements.

2.3 Mesh Quality Assessment

The accuracy and stability of the numerical solution are directly influenced by the quality of the computational mesh. In this study, three primary metrics were utilized to evaluate the mesh: Element Quality, Aspect Ratio, and Skewness.

- **Element Quality:** This composite metric evaluates the shape of the element on a scale from 0 to 1, where 1 represents a perfect equilateral cell and 0 indicates a degenerate cell with zero or negative volume. For 3D elements, it is calculated based on the ratio of the volume to the square root of the cube of the sum of the square of the edge lengths.
- **Aspect Ratio:** Defined as the ratio of the longest edge length to the shortest edge length of an element. High aspect ratios can lead to numerical instability, particularly in regions with high flow gradients; however, values observed within the boundary layer (inflation layers) remain acceptable for the selected numerical setup.
- **Skewness:** This metric measures how close an element is to its ideal equilateral form. A skewness of 0 represents an ideal cell, while values approaching 1 indicate highly distorted elements that may hinder convergence.

The statistical distribution of these metrics is illustrated in Figure 2.2. The mean values recorded—0.77 for Element Quality, 1.83 for Aspect Ratio, and 0.22 for Skewness—demonstrate a high-quality discretization suitable for steady species transport analysis.



(a) Element Quality (b) Aspect Ratio (c) Skewness

Figure 2.2: Mesh quality histograms: (a) Element Quality distribution (Mean: 0.77); (b) Aspect Ratio distribution (Mean: 1.83); (c) Skewness distribution (Mean: 0.22).

The statistical distributions of Element Quality, Aspect Ratio, and Skewness are illustrated in Figure 2.2. The corresponding mean values are 0.77, 1.83, and 0.22, respectively, indicating good overall mesh quality.

Although the maximum aspect ratio reaches 31.65 near the walls due to the boundary-layer inflation layers, these extreme values are highly localized. In the critical flow region, most elements retain low skewness and acceptable shape characteristics, which helps reduce numerical diffusion during the steady species transport simulation.

Chapter 3

Numerical Setup and Boundary Conditions

3.1 Governing Equations

The fluid flow is governed by the steady-state 3D Reynolds-Averaged Navier-Stokes (RANS) equations. Since the simulation is performed under steady conditions, the time-dependent terms are neglected. The mass conservation and momentum equations are expressed as follows:

$$\nabla \cdot (\rho \mathbf{u}) = 0 \tag{3.1}$$

$$\nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot (\tau) + \rho \mathbf{g} \tag{3.2}$$

For the pollutant dispersion, a species transport model without chemical reactions was employed. The local mass fraction of the pollutant (SF_6), denoted as Y_i, is predicted through the solution of a convection-diffusion equation:

$$\nabla \cdot (\rho \mathbf{u} Y_i) = -\nabla \cdot \mathbf{J}_i + S_i \tag{3.3}$$

where $\mathbf{J}_i$ is the diffusion flux of the species.

3.2 Material Properties and Species Transport

The dispersion study involves a mixture of air and Sulfur hexafluoride (SF_6) as a tracer gas. To accurately simulate the buoyancy effects, the following properties were implemented:

- **Tracer Gas (SF_6):** The molecular weight is set to 146.06 kg/kmol, making it approximately five times heavier than air. The density is modeled using the ideal-gas law, and the viscosity is defined as 1.53×10^{-5} Pa·s.
- **Mixture Properties:** The background air viscosity is 1.7894×10^{-5} kg/m·s. The mass diffusivity for the SF_6-air mixture is specified as 9.5×10^{-6} m^2/s.
- **Turbulent Schmidt Number:** Following the standard recommendations for atmospheric dispersion, the Turbulent Schmidt Number (Sc_t) is set to 0.7.

3.3 Boundary Conditions and UDF

To simulate a realistic Atmospheric Boundary Layer (ABL) under neutral stability conditions, the inlet velocity, turbulent kinetic energy (k), and dissipation rate (ε) were prescribed using a User-Defined Function (UDF). This approach ensures an equilibrium ABL, preventing the artificial decay of turbulence as the flow traverses the computational domain. The vertical profiles are defined by the following analytical expressions:

$$u(z) = 4.7\left(\frac{z}{H}\right)^{0.3} \tag{3.4}$$

$$k = \frac{u_*^2}{\sqrt{C_\mu}}\left(1 - \frac{z}{\delta}\right) \tag{3.5}$$

$$\varepsilon = \frac{u_*^3}{\kappa z}\left(1 - \frac{z}{\delta}\right) \tag{3.6}$$

Where the physical parameters and constants are defined as follows:

- $H = 18$ m is the reference building height.
- $\delta = 144$ m is the boundary layer depth, scaled to the full-size geometry.
- $u_* = 0.54$ m/s is the friction velocity.
- $\kappa = 0.4$ is the von Kármán constant.
- $C_\mu = 0.09$ is the standard empirical constant for the $k - \varepsilon$ model.

Listing 3.1: UDF

```
#include "udf.h"

#define U_STAR 0.54
#define DELTA 144.0
#define KAPPA 0.4
#define CMU 0.09

DEFINE_PROFILE(velocity_inlet, thread, position)
{
  real x[ND_ND], z;
  face_t f;
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z < 0.01) z = 0.01;
    F_PROFILE(f, thread, position) = 4.7 * pow(z / 18.0, 0.3);
  }
  end_f_loop(f, thread)
}

DEFINE_PROFILE(k_profile, thread, position)
{
  real x[ND_ND], z;
  face_t f;
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z > DELTA) z = DELTA;
    F_PROFILE(f, thread, position) = (pow(U_STAR, 2) / sqrt(CMU)) *
        (1.0 - z/DELTA);
  }
  end_f_loop(f, thread)
}

DEFINE_PROFILE(eps_profile, thread, position)
{
  real x[ND_ND], z;
  face_t f;
```

```
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z < 0.1) z = 0.1;
    if (z > DELTA) z = DELTA * 0.99;

    F_PROFILE(f, thread, position) = (pow(U_STAR, 3) / (KAPPA * z)) *
       (1.0 - z/DELTA);
  }
  end_f_loop(f, thread)
}
```

The boundary conditions were defined to accurately replicate the atmospheric boundary layer and the specific pollutant emission source:

- **Main Inlet:** A velocity-inlet profile implemented via UDF with $u_* = 0.54$ m/s and a boundary layer depth $\delta = 144$ m (scaled to full-size geometry).
- **Pollution Source:** A *mass-flow-inlet* was used for the SF_6 release with a rate of 0.00015 kg/s. The turbulence parameters at the source were set to a 2% intensity and a turbulent viscosity ratio of 2.5.
- **Outlet:** A pressure-outlet boundary condition with radial equilibrium pressure distribution was applied to allow the flow to leave the computational domain.
- **Walls:** No-slip conditions were applied to all building and ground surfaces.

3.4 Solver Configuration

The numerical solution was obtained using the following settings in ANSYS Fluent:

- **Solver Type:** Pressure-based steady-state solver.
- **Pressure-Velocity Coupling:** SIMPLE.
- **Gravity:** Enabled in the negative Z-direction (-9.81 m/s^2).
- **Turbulence Model:** Standard $k - \varepsilon$ model with $C_\mu = 0.09$, $C_{1\varepsilon} = 1.44$, and $C_{2\varepsilon} = 1.92$.
- **Discretization Schemes:** Second-order upwind for momentum and species transport.

3.5 Solver Stability and Convergence

To ensure stable convergence and prevent numerical oscillations during the steady-state iterations, the Under-Relaxation Factors (URF) were optimized. Higher values were assigned to the species and energy equations to prioritize their convergence after the flow field became partially developed.

Table 3.1: Applied Under-Relaxation Factors

Variable	**Factor Value**
Pressure	0.3
Momentum	0.7
Turbulent Kinetic Energy (k)	0.8
Dissipation Rate (ε)	0.8
Air (Mass Fraction)	0.9
SF_6 (Mass Fraction)	0.9
Energy	1.0

The solution was considered converged when the scaled residuals for all equations reached a threshold of 1×10^{-5}. In addition to residuals, the mass fraction of SF_6 at the outlet was monitored to ensure it reached a steady-state value.

Chapter 4

Results and Discussion

4.1 Flow Field in the Street Canyon

Sample: The airflow inside the street canyon is characterized by a dominant recirculation vortex. Due to the interaction between the approaching wind and the building geometry, the flow rotates inside the canyon and redistributes the pollutant released at street level. This recirculation structure plays a key role in pollutant transport and strongly influences the concentration field near the walls.

4.2 Mean Concentration Distribution of SF_6

Sample: The concentration contours show that the pollutant is not distributed uniformly inside the canyon. Higher concentration levels are observed in weakly ventilated regions, especially near the wall affected by the main recirculation motion. In contrast, lower concentrations appear in regions where the airflow promotes pollutant removal from the canyon.

4.3 Concentration Profiles at Selected Locations

Sample: The vertical concentration profiles provide a more detailed view of pollutant distribution inside the canyon. The highest values are observed in the central region, where the effect of recirculation is strongest. As the height increases, the concentration generally decreases, indicating gradual mixing with the outer flow above the canyon.

4.4 Validation and Numerical Convergence

Sample: The solution was considered converged when the scaled residuals dropped below the prescribed threshold of 10^{-5}. In addition, the monitored outlet values approached a stable level, indicating numerical consistency of the simulation. Where reference data are available, the predicted concentration trends show reasonable agreement with the benchmark results.